# Enhanced-YOLOvDP: A Lightweight Real-Time Detection Model for UAV-Based Plant Disease Monitoring on Edge Platforms

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**Abstract.** Early detection of plant diseases is essential for maintaining agricultural productivity, particularly in large-scale farming environments where manual inspection is inefficient and labor intensive. This study proposes an Enhanced-YOLOvDP model for UAV-based plant disease detection with a focus on achieving a balance between detection accuracy and real-time inference performance. The proposed model simplifies the original YOLOvDP architecture by replacing computationally expensive modules and reducing redundant feature fusion layers, thereby improving computational efficiency while maintaining reliable detection capability. A UAV-based PDT dataset containing 5,670 annotated images is used for model training and evaluation. Experimental results show that the proposed model achieves a precision of 0.887, recall of 0.831, and mAP@0.5 of 0.901 while reaching an inference speed of 116.2 FPS. After INT8 post-training quantization, the model maintains stable detection performance with improved runtime efficiency. Hardware deployment experiments demonstrate that the model achieves up to 139.89 FPS on the ZCU102 FPGA platform, significantly outperforming CPU-based edge devices. These results confirm that the proposed approach is well suited for real-time UAV-based agricultural monitoring and edge AI deployment.

**Keywords:** Plant Disease Detection, UAV-based Monitoring, YOLO Object Detection, Edge AI, FPGA Acceleration, Quantized Neural Networks, Precision Agriculture

## 1 Introduction

The applications of Artificial Intelligence in agriculture plays a critical role in supporting the sustainable development of Vietnam's economy [1], [2]. However, large scale farming systems require substantial effort to ensure continuous monitoring of crop health and disease conditions in order to maintain productivity

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and operational efficiency. Effective supervision of plant growth and early disease detection are essential for minimizing yield loss and ensuring stable agricultural output [3].

Traditional disease detection methods based on manual observation present several limitations. They are time consuming, labor intensive, and heavily dependent on skilled experts [3]. For large agricultural areas, manual inspection becomes impractical, resulting in delayed disease detection, reduced diagnostic accuracy, and increased risk of crop quality degradation and productivity loss [4]. These constraints highlight the need for scalable and automated monitoring solutions.

In recent years, Artificial Intelligence techniques, particularly Convolutional Neural Networks(CNN), have been widely adopted for plant disease detection through image analysis [5]. The CNN based models have demonstrated strong capability in learning discriminative visual features, enabling accurate classification and localization of infected regions. In addition, Transformer based architectures have shown promising performance in modeling complex spatial relationships within images, further improving diagnostic accuracy [6], [7]. Nevertheless, most existing studies focus on images captured in localized and small scale settings, typically using smartphone cameras or sensors. Such approaches are not suitable for large scale farming environments where extensive spatial coverage is required.

To overcome these limitations, the use of Unmanned Aerial Vehicles (UAV) has emerged as a promising direction for large scale crop monitoring [8]. UAV based remote sensing enables rapid coverage of extensive agricultural fields, providing flexible flight altitude and high spatial resolution imagery. This capability significantly enhances both spatial coverage and temporal monitoring frequency, which are crucial for early disease detection in precision agriculture systems. Compared with manual scouting or fixed ground sensors, UAV platforms offer superior scalability and operational efficiency.

Furthermore, the integration of AI with embedded systems has enabled intelligent processing directly on UAV platforms. UAV imagery combined with machine learning models allows automated disease identification, reducing human intervention while improving consistency and repeatability. With recent advances in embedded computing and lightweight deep learning architectures, edge devices such as Raspberry Pi or NVIDIA Jetson based modules can perform near real time inference during flight [9]. This edge based deployment balances latency, computational efficiency, and energy consumption, making it suitable for practical large scale agricultural monitoring systems [10].

The integration of edge computing into UAV-based crop disease detection has gained increasing attention due to the need for low-latency and energy-efficient processing in large-scale agricultural environments [11]. Traditional cloud-based processing introduces significant communication overhead, increased latency, and potential connectivity issues, particularly in rural or remote farming areas. To address these limitations, edge devices deployed directly on UAV platforms enable on-board inference, reducing reliance on network infrastructure and allowing real-time decision-making during flight [12]. Recent surveys highlight that edge-enabled UAV systems significantly improve responsiveness and operational autonomy in precision agriculture applications [13].

Real-time inference is particularly critical in UAV disease monitoring because UAVs continuously move across large farming areas, and high-resolution imagery must be processed promptly to enable adaptive flight control, immediate disease localization, or early intervention strategies. Studies emphasize that reducing end-to-end latency is essential for maintaining system stability and responsiveness in mobile edge computing scenarios [9]. In agricultural UAV applications, latency requirements are often constrained to tens of milliseconds per frame, especially for detection-based models operating on streaming video data [11]. To meet these constraints, lightweight deep learning architectures such as MobileNet and YOLO-nano variants have been adopted to reduce computational complexity while maintaining acceptable accuracy [12], [1] . Furthermore, model compression techniques including quantization and pruning are widely reported to reduce memory footprint and dynamic power consumption without severe performance degradation. Hardware-aware optimization and low-bit quantization significantly enhance the feasibility of deploying deep neural networks on resource-constrained edge devices [14], [15]. Significant challenges remain in deploying AI models for UAV based disease detection. Many existing approaches primarily emphasize improving detection accuracy, while limited attention is given to real time performance and computational constraints. As a result, models that perform well in offline evaluation often fail to meet the latency and resource requirements necessary for practical UAV operations.

This study proposes a joint optimization strategy that simultaneously considers model architecture and inference latency to achieve a balanced trade off between detection accuracy and real time performance for practical UAV deployment. An Enhanced-YOLO based detection model is designed to maintain reliable localization and classification of plant disease regions while ensuring fast inference. The model is tailored to process large scale UAV imagery efficiently, enabling automated monitoring over extensive farmland and supporting early disease intervention with reduced reliance on manual inspection. To enable deployment on resource constrained hardware platforms, the model is further optimized through architectural simplification and post training quantization. These techniques reduce computational complexity, memory usage, and power consumption while preserving acceptable detection performance. The optimized model is then deployed on an embedded hardware board and evaluated under practical operating conditions, with emphasis on detection accuracy, inference speed in frames per second, and overall system efficiency.

## 2 Methodology

### 2.1 YOLOvDP overview

YOLOvDP is a customized lightweight object detection framework designed for UAV-based agricultural disease monitoring in large-scale farmland environments. The model is built upon the YOLOv5 architecture, which has demonstrated strong real-time detection performance in various vision tasks [16]. To enhance detection capability in aerial imagery, YOLOvDP integrates data-driven anchor generation based on K-means clustering, a strategy originally adopted in early YOLO versions to better align anchor boxes with dataset distributions [17].

To address the challenges of small and densely distributed targets in UAV scenes, the model incorporates an adaptive large-scale selective kernel mechanism inspired by selective kernel networks [18], enabling dynamic receptive field adjustment for improved contextual feature extraction. Computational efficiency is further improved through the use of Ghost convolution, which reduces parameter redundancy while maintaining representational power [19]. In addition, multi-scale feature fusion is achieved using an FPN and PAN structure to enhance robustness across object scales [20], [21].

Through these architectural enhancements, YOLOvDP achieves a balanced trade-off between detection accuracy and computational efficiency, making it suitable for real-time deployment on resource-constrained UAV platforms for precision agriculture applications [22].

### 2.2 Dataset

The PDT dataset is developed to support UAV-based detection of unhealthy pine trees in large-scale agricultural and forestry environments [22]. It is designed to meet the requirements of high spatial coverage and accurate target localization from aerial perspectives. Data collection was conducted using a DJI-ChanSi L2 UAV mapping system equipped with high-resolution imaging, LiDAR, and precise navigation modules, enabling reliable acquisition under different flight altitudes and environmental conditions. Disease regions are identified based on visual differences between healthy and infected trees, allowing models to learn discriminative features across varying UAV viewpoints.

The dataset construction followed a structured pipeline including data acquisition, preprocessing, annotation, and quality control. A total of 105 ultra-high-resolution images and corresponding 3D point cloud data were collected at a flight altitude of 150 meters. Due to the large image size, raw images were systematically processed into smaller patches suitable for deep learning-based object detection. 640x640 pixel size images is processed with sliding windowns to create 54 images per large image size. As shown in Table 1, this study contains a total of 5,670 images collected for plant disease detection and analysis. The training set contains 4,536 images. The validation set consists of 567 images. The test set includes 567 images and is used only for the final evaluation of the trained model.

**Table 1.** Distribution of Images in the PDT Dataset

| | Totl Images | Train | Test | Validation |
|---|---|---|---|---|
| Number of Images Used | 5,670 | 4,536 | 567 | 567 |

# 3 Model Development

## 3.1 Enhanced-YOLOvDP model

The design of the Enhanced-YOLOvDP model is motivated by the practical requirements of UAV-based plant disease detection, where real-time inference and computational efficiency are as critical as detection accuracy. In real-world agricultural monitoring scenarios, UAV platforms operate under strict constraints on processing power, memory, and energy consumption. Therefore, the original YOLOvDP architecture, while effective in feature representation, requires further simplification to achieve a better balance between accuracy and inference speed. Figure 1 shows the Enhanced-YOlOvDP. Compared with the original YOLOvDP architecture, the enhanced version prioritizes computational efficiency by replacing the Large-Scale Selective Kernel module with the standard C3 block. While the LSK module improves feature representation through enlarged receptive fields and attention-based selection, it also increases parameter count and floating-point operations, which can limit real-time performance. In contrast, the C3 block adopts cross-stage partial connections combined with conventional convolution layers, offering a more lightweight yet sufficiently expressive feature extraction mechanism. This modification reduces computational burden while maintaining stable detection capability.

In addition, the neck network is streamlined by removing the P3 feature fusion branch. In the original configuration, the P3 layer preserves high-resolution features and enhances the representation of relatively larger disease regions through multi-scale fusion. However, this process requires additional upsampling, concatenation, and convolution operations, leading to higher memory usage and increased inference latency. By eliminating the P3 pathway, the Enhanced-YOLOvDP reduces redundant feature aggregation and shortens the information flow within the neck. Consequently, fewer feature maps participate in multi-scale fusion, resulting in lower memory access and computational cost.

Although this simplification slightly reduces the model's ability to capture large-scale targets due to diminished high-resolution semantic integration, the impact on overall accuracy is minimal because most objects in the PDT dataset are small to medium in size. Experimental results demonstrate that these architectural adjustments significantly improve inference speed, achieving higher frames per second while preserving competitive detection performance. Therefore, the Enhanced-YOLOvDP offers a more practical solution for real-time deployment on resource-constrained UAV platforms.

## 3.2 Data Pre-Processing

To enable effective training of mainstream object detection models on ultra–high-resolution UAV imagery, a structured multi-stage preprocessing pipeline was developed for the PDT dataset. The original aerial images, with a resolution of $5472 \times 3648$ pixels, were defined as high-altitude (LH) samples. Due to their large spatial dimensions, directly feeding these images into deep neural networks would

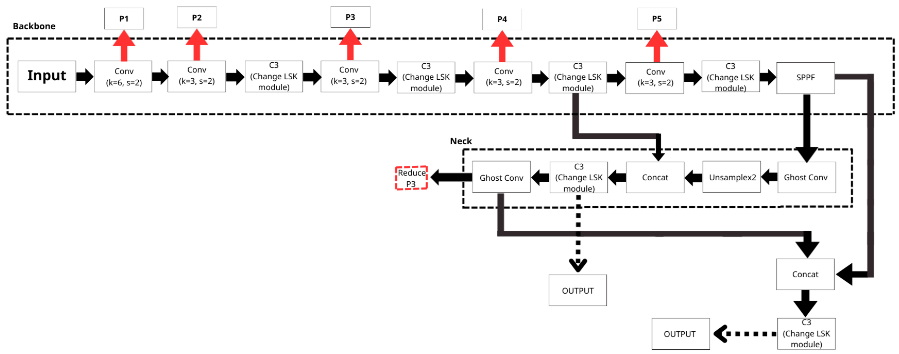


**Fig. 1.** Architecture of the proposed YOLOvDP model.

result in excessive computational and memory demands. Therefore, a sliding window approach was employed to partition each LH image into manageable patches. Fixed-size crops of 640 $\times$ 640 pixels were generated with a stride of 630 $\times$ 630 pixels, introducing slight overlap between adjacent patches to preserve contextual continuity and prevent information loss near boundaries. On average, approximately 54 low-altitude (LL) sub-images were produced from each LH image.

Both target-containing patches and background-only samples were retained during preprocessing. Including background images enables the model to learn discriminative representations between diseased trees and visually similar surroundings, thereby reducing false positives during inference. An initial subset of LL images was manually annotated using Labelme and used to train a YOLOv5s model for 300 epochs. The trained detector was then integrated into a Human-in-the-loop framework to automatically generate preliminary bounding boxes for the remaining LL images, using a confidence threshold of 0.25 and a non-maximum suppression IoU threshold of 0.45.

All auto-generated annotations were subsequently reviewed and refined through manual verification to ensure labeling accuracy and completeness. The finalized LL dataset was exported in YOLO formats and divided into training, validation, and testing subsets with an 8:1:1 ratio. These refined LL annotations were further utilized to assist annotation of the original LH images using the YOLOvDP model, followed by an additional Human-in-the-loop verification stage. This dual-resolution annotation strategy enhances both labeling efficiency and detection robustness, enabling the PDT dataset to support high-density and high-precision UAV-based plant protection tasks.

## 4 Result

The training dynamics over 50 epochs are analyzed through the progression of Precision, Recall, and mAP@50. During the early training phase, all metrics

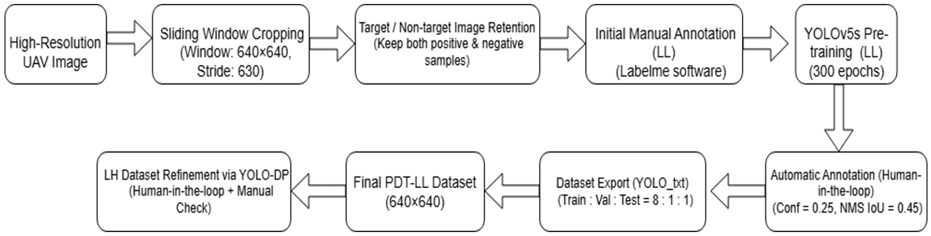


**Fig. 2.** Data Pre-Processing diagram.

exhibit a rapid increase, indicating efficient feature extraction and effective parameter initialization. As illustrated in Fig. 3(a), the growth of the performance of the YOLOvDP baseline model begins to moderate after approximately the 10th epoch, suggesting that the model enters a stable optimization stage with progressively refined bounding box regression and classification performance. In the final training stage, Precision and Recall converge to approximately 0.87 and 0.85, respectively, while mAP@50 stabilizes near 0.93. The smooth convergence trend and absence of noticeable fluctuations indicate stable learning behavior and reliable detection capability.

The Enhanced-YOLOvDP architecture in Fig. 3 (b) shows faster convergence and better final performance. The mAP@50 exceeds 0.90 around the 15th epoch, earlier than the baseline. In later epochs, Precision and Recall stabilize at about 0.90 and 0.86, respectively, while mAP@50 approaches 0.94. This stable convergence indicates that the architectural improvements enhance training efficiency and detection robustness without causing overfitting.

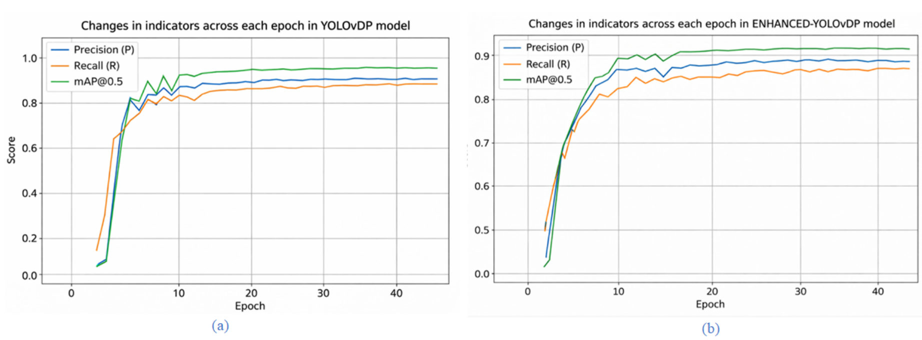


**Fig. 3.** Training performance across each epoch in (a) YOLOv-DP model (b) Enhanced-YOLOv-DP.

Table 2 shows that the improved model increases FPS but slightly reduces mAP@0.5:0.95. This decrease mainly results from two architectural changes: removing the P3 feature scale and replacing standard convolution blocks in the neck. The P3 layer normally captures high-level semantic features with a large receptive field, which helps precise localization under strict IoU thresholds. Its removal reduces the model's ability to preserve global context and refine bounding boxes.

Despite this, Enhanced-YOLOvDP achieves a good balance between accuracy and real-time performance on the PDT dataset. It reaches a precision of 0.887, recall of 0.831, and F1-score of 0.858, outperforming the original YOLOvDP (0.862, 0.794, 0.827). The model also achieves mAP@0.5 of 0.901, higher than YOLOvDP, YOLOv5s, and YOLOv5n, while remaining competitive with newer models such as YOLOv8n and YOLOv11n. With 6.09M parameters and 12.8 GFLOPs, it remains lightweight and achieves the fastest inference speed of 116.2 FPS, making it suitable for real-time plant disease detection on edge devices.

**Table 2.** Pre-quantized Performance Comparison of Detection Models on the PDT Dataset

| Model | Precision | Recall | F1 | mAP@0.5 | mAP@0.5:0.95 | Params | GFLOPs | FPS |
|---|---|---|---|---|---|---|---|---|
| Enhanced-YOLOvDP (our) | 0.887 | 0.831 | 0.858 | 0.901 | 0.603 | 6,099,030 | 12.8 | 116.2 |
| YOLOvDP | 0.862 | 0.794 | 0.827 | 0.863 | 0.571 | 4,993,774 | 11.1 | 51.02 |
| YOLOv5s | 0.877 | 0.815 | 0.845 | 0.882 | 0.605 | 7,012,822 | 15.8 | 81.97 |
| YOLOv5n | 0.873 | 0.808 | 0.839 | 0.880 | 0.588 | 1,760,518 | 4.1 | 80.00 |
| YOLOv8n | 0.8721 | 0.8455 | 0.8586 | 0.9247 | 0.6396 | 3,011,043 | 8.19 | 78.46 |
| YOLOv11n | 0.875 | 0.8403 | 0.8573 | 0.9234 | 0.6394 | 2,590,035 | 6.44 | 67.98 |

Table 3 shows that both models were evaluated on the same dataset with 12,655 ground truth boxes, ensuring a fair comparison. The original YOLOvDP achieves slightly higher localization performance, with Max IoU of 0.989 and Mean IoU of 0.8257, compared to 0.9859 and 0.8223 for Enhanced-YOLOvDP. Both models report a Min IoU of 0, indicating a few missed or poorly localized objects under challenging conditions. The difference is minimal, showing that Enhanced-YOLOvDP maintains comparable localization accuracy despite architectural simplifications.

**Table 3.** IoU and Detection Statistics Comparison between YOLOvDP and Enhanced-YOLOvDP

| Metric | YOLOvDP | Enhanced-YOLOvDP |
|---|---|---|
| Total GT boxes | 12,655 | 12,655 |
| Max IoU | 0.989 | 0.9859 |
| Min IoU | 0 | 0 |
| Mean IoU | 0.8257 | 0.8223 |

To support efficient deployment in real-world agricultural monitoring, a quantized version of the Enhanced-YOLOvDP is developed using a post-training quantization strategy. The model is first trained in full precision (FP32) on the Kaggle platform, and the best weights are saved as a PyTorch checkpoint (best.pt). For deployment flexibility, the trained model is exported to ONNX format, preserving the original computational graph and numerical accuracy.

Using the OpenVINO toolkit, the FP32 ONNX model is converted into an INT8 OpenVINO Intermediate Representation (IR) [23]. This quantization process significantly reduces memory usage and computational cost. Compared to the FP32 baseline, the INT8 model achieves lower latency and higher inference speed as shown in Table 4, making it more suitable for real-time UAV-based plant disease detection under hardware constraints.

**Table 4.** Post-quantized Performance Comparison using GTX 1650 GPU

| Model | Precision | Recall | F1 | mAP@0.5 | mAP@0.95 | Params | FPS | Layers |
|---|---|---|---|---|---|---|---|---|
| Enhanced-YOLOvDP | 0.8362 | 0.7948 | 0.8150 | 0.8771 | 0.5256 | 6,136,316 | 27.11 | 202 |
| YOLOv-DP | 0.8316 | 0.8073 | 0.8193 | 0.8843 | 0.5523 | 5,094,674 | 10.58 | 309 |
| YOLOv5s | 0.8131 | 0.8056 | 0.8093 | 0.8667 | 0.4914 | 7,123,564 | 16.19 | 236 |
| YOLOv5n | 0.7873 | 0.7734 | 0.7803 | 0.8398 | 0.4455 | 1,866,508 | 43.17 | 263 |
| YOLOv8n | 0.8727 | 0.8439 | 0.8581 | 0.9240 | 0.6410 | 3,005,843 | 16.07 | 233 |
| YOLOv11n | 0.8728 | 0.8431 | 0.8577 | 0.9240 | 0.6400 | 2,582,347 | 15.87 | 320 |

Table 4 compares post-quantization performance of different detection models on a GTX 1650 GPU. Enhanced-YOLOvDP maintains competitive accuracy with a precision of 0.8362, recall of 0.7948, and F1-score of 0.8150. It achieves mAP@0.5 of 0.8771 and mAP@0.95 of 0.5256, showing that localization capability is largely preserved after quantization.

Although YOLOv8n and YOLOv11n reach higher accuracy (mAP@0.5 $\approx$ 0.924), Enhanced-YOLOvDP offers a balanced trade-off between accuracy and complexity with 6.13M parameters and 202 layers. In terms of speed, it achieves 27.11 FPS, significantly faster than YOLOvDP (10.58 FPS) and YOLOv5s (16.19 FPS), though slower than YOLOv5n (43.17 FPS). These results indicate that Enhanced-YOLOvDP maintains stable detection performance while improving runtime efficiency, making it suitable for real-time plant disease detection.

Figure 4 illustrates a trade-off between the original YOLOvDP and Enhanced-YOLOvDP in qualitative detection results. In Fig. 4 (a), YOLOvDP provides more precise localization of diseased regions, especially in dense vegetation where targets are small and clustered. The bounding boxes are tighter and better aligned, indicating stronger feature extraction and spatial sensitivity.

In contrast, Fig. 4 (b) shows that Enhanced-YOLOvDP prioritizes computational efficiency and real-time inference. This leads to a slight reduction in accuracy, with some small or dense regions occasionally missed and bounding boxes less refined in complex backgrounds, along with a few false positives. These effects result from architectural simplifications designed to reduce complexity and accelerate inference. Despite the minor accuracy loss, Enhanced-YOLOvDP achieves significantly higher inference speed, making it more suitable for real-time deployment on resource-constrained UAV platforms.

(a) (b)

**Fig. 4.** Image cropped from the detection area of (a) YOLOvDP model (b) Enhanced-YOLOvDP.

Figure 5 shows that the Enhanced-YOLOvDP deployed on the ZCU102 platform accurately detects diseased regions in aerial crop images. The red bounding boxes highlight multiple affected areas even in complex backgrounds where disease spots are small and scattered. The model also demonstrates effective multi-scale detection by identifying both large clusters and small isolated regions. The well-aligned bounding boxes and low number of false detections confirm that the model performs reliably on the ZCU102 platform and is suitable for real-time UAV-based plant disease monitoring. The deployment results show that the ZCU102 platform achieves higher detection accuracy and significantly better inference performance than the Raspberry Pi 4 running the quantized model with OpenVINO. ZCU102 reaches a precision of 0.8622, recall of 0.8514, and mAP@0.5 of 0.8884, slightly higher than Raspberry Pi 4 (precision 0.8382, recall 0.8029, mAP@0.5 of 0.8818). This improvement is mainly due to the Vitis-AI quantization and compilation process optimized for the ZCU102 DPU architecture, which preserves numerical precision more effectively than CPU-based OpenVINO inference on Raspberry Pi 4.

In terms of efficiency, ZCU102 achieves 61.62 FPS with a single inference flow and up to 139.89 FPS with four parallel flows, while Raspberry Pi 4 reaches only 0.85 FPS. These results demonstrate that the DPU-based FPGA platform is well suited for real-time inference and parallel processing, making ZCU102 a more effective solution for edge-based agricultural monitoring applications.

**Fig. 5.** Two different image detection results deployed on ZCU102 board using Enhanced-YOLOv-DP.

**Table 5.** Detection Performance Metrics of Enhanced-YOLOvDP on ZCU102 and Raspberry Pi 4

| Platform | Precision | Recall | mAP@0.5 | mAP@0.5:0.95 | FPS (1 flow) | FPS (4 flows) |
|---|---|---|---|---|---|---|
| ZCU102 | 0.8622 | 0.8514 | 0.8884 | 0.5723 | 61.6228 | 139.891 |
| Raspberry Pi 4 | 0.8382 | 0.8029 | 0.8818 | 0.5327 | 0.85 | NA |

## 5 Conclusion

This study presents an Enhanced-YOLOvDP model designed for real-time UAV-based plant disease detection in large-scale agricultural environments. By simplifying the original YOLOvDP architecture and optimizing feature fusion mechanisms, the proposed model achieves an effective balance between detection accuracy and computational efficiency. Experimental results on the PDT dataset demonstrate that the Enhanced-YOLOvDP model achieves competitive detection performance while significantly improving inference speed compared with several lightweight detection models. To enable practical deployment on edge platforms, post-training INT8 quantization is applied to reduce computational complexity and memory requirements. The quantized model maintains stable detection accuracy while improving inference efficiency. Hardware evaluation further shows that the FPGA-based ZCU102 platform significantly outperforms CPU-based embedded systems such as Raspberry Pi 4 in both detection accuracy and runtime performance. The system achieves up to 139.89 FPS with parallel processing, demonstrating strong suitability for real-time UAV applications. the proposed framework provides an effective solution for automated crop disease monitoring using UAV imagery and edge AI. Future work will focus on improving multi-scale detection capability, further reducing model complexity, and integrating the system with fully autonomous UAV monitoring pipelines for large-scale precision agriculture

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